

# Parth Patel

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## EDUCATION

**Georgia Institute of Technology, College of Engineering**  
**Bachelor of Science in Mechanical Engineering**

**Atlanta, Georgia**  
*Expected Graduation: May 2026*

- Zell Miller Scholar, Lead Engineer at Wreck Racing, Aerodynamics at Supersonics Club, Fall 2025 Dean's List (**3.4 GPA**)

## EXPERIENCE

**Yellow Jacket Space Program**  
**Propulsion Engineer**

**Atlanta, Georgia**  
*Aug 2025 – Present*

- Developed a LOX coaxial swirl injector achieving 92%  $c^*$ -efficiency through iterative and hot fire testing campaigns
- Optimized 18-element flight hardware from single element prototypes by conducting thermal and discharge modeling

**Summit Capital Investment Group LLC**  
**Real Estate Developer**

**Lawrenceville, Georgia**  
*Mar 2022 – Aug 2025*

- Managed a \$30M public-private development, building LEED Certified housing for essential workers in Vail, CO
- Directed critical infrastructure integration, validating schematics against strict energy and environmental constraints.

**Deloitte Government and Public Services**  
**Technology Strategy Summer Analyst**

**Atlanta, Georgia**  
*May 2022 – Aug 2022*

- Assisted a DoD/intelligence agency in creating new SOPs for talent acquisition to streamline federal hiring initiatives
- Developed a strategic playbook and multi-phase timeline to accelerate the global scaling of a new engineering team

**Siemens Healthineers**

**Cary, North Carolina**  
*Jan 2022 – April 2022*

**Research and Development Intern**

- Worked on the development of a lab diagnostics product (Siemens Attelica) for faster turnaround in immunoassay testing
- Tracked the deployment and logistics of this solution being shipped to over 70 countries utilizing Tableau and PowerBI

**McKinsey & Company**

**Atlanta, Georgia**  
*Jan 2021 – Dec 2021*

**Knowledge Analyst (Design-to-Value Lab)**

- Conducted product teardowns to identify optimization opportunities with an average of 25% reduction in COGS for clients
- Generated product ideas for clients in consumer and industrial sectors by applying comparative design and benchmarking

**The Home Depot**

**Atlanta, Georgia**  
*Jan 2020 – Dec 2020*

**Global Brand and Product Development Engineer Co-op**

- Worked alongside Shaw and Mohawk Ind. to develop new test methodologies for carpets and flooring that are overall 22% more time efficient during onboarding of new products, resulting in \$32 million saved in yearly testing costs
- Conducted product quality assessments and updated Flex PLM workflow to ensure compliance with SOPs and contracts

## PROJECTS

**Create-X at Georgia Tech**

**Atlanta, Georgia**  
*Aug 2025 – Present*

**Idea-to-Prototype and Capstone Design**

- Leading the design and integration of an autonomous counter-UAS swarm system for threat detection and kinetic intercept
- Developing a ground station using dual sensing (computer vision + doppler radar) to guide the drone swarm against threats

**Georgia Tech Guggenheim School of Aerospace Engineering**

**Atlanta, Georgia**  
*May 2024 – Aug 2024*

**Wind Engineering (AE 4701)**

- Optimized the rotor and blade design of a 1 MW wind turbine based in Colorado using WT\_PERF to maximize power
- Conducted ANSYS CFD and FEA simulations to map aerodynamic pressure distributions and validate performance

**InspireNC (501c3)**

**Raleigh, North Carolina**  
*July 2017 – Dec 2020*

**Co-Founder and Director**

- Established a 9,000 square foot community STEAM workspace and peer-mentor system in the Research Triangle Park Area
- In year 1, raised over \$750,000 through grants, advocacy, and crowdfunding and increased membership from 0 to 75

**Georgia Tech Lab for Ultrasonic Non-destructive Testing**

**Metz, France**  
*May 2019 – Aug 2019*

**Research Assistant to Dr. Declercq**

- Tested metal plates for ship hull defect detection using acoustic transducers to process ultrasonic waves in MATLAB
- Assisted in proposal writing for securing a \$350,000 ONR grant for defect detection in Navy warships

## SKILLS

**Computer Aided Design:** Solidworks Suite, Siemens NX, Autodesk (Inventor and Fusion 360), ANSYS, FEA, FMEA, CFD

**Technical:** MATLAB, Simulink, VPython, BASIC, EES, LabVIEW, Maple, JIRA, PowerBI/PowerApps, Tableau, Flex PLM

**Creative:** Microsoft Office, Vicon Motion Capture, FL Studio, Final Cut Pro X, Cyberlink Suite, Adobe CC Suite, GD&T

**Manufacturing:** CNC Mill, 3-D Printer, Laser cutter, Water Jet Cutter, Wire EDM, Saws, Drill Presses, Vertical Mill, Lathe

**Languages:** Gujarati (Fluent second language), Hindi (Advanced), French (Working Proficient), Spanish (Elementary)

**Additional Skills:** Project and Risk Management, Data and Statistical Analyzing, Human-Computer Interaction, Agile, CRM